Noise Removal versus Signal Loss in Network Filtering of Daily GNSS Time Series

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Abstract

Daily GNSS position time series are routinely filtered to suppress noise that is shared across stations, but the same filters can remove real transient deformation, because slow slip, afterslip and loading signals are spatially coherent too. Common-mode filters have been judged mainly by how much scatter they remove, a criterion that also rewards removing signal. Where signal preservation has been examined, it has been inferred from the consistency of noisy observed offsets or tested with synthetic signals in synthetic noise, and never as a function of the size of a transient relative to the reach of the filter. A recent graph-neural-network study in Cascadia showed that predicting each station only from stations more than 400 km away preserves slow slip, but chose that distance empirically for one region. Here we propose a signal-loss audit of network filtering. We plant transients of known amplitude, width and duration in the real noise of 1,251 continuous stations in and around California, and measure for each filter the fraction of each transient it retains, the spurious signal it creates at nearby stations, and the noise it removes. We audit three families under one leakage-controlled protocol: classical filters that use the target station itself (stacking, robust median imaging and regional principal component analysis), filters that exclude stations within a fixed distance (the 400 km design), and leave-one-out reconstruction models with an exclusion radius varied systematically, from weighted means to a graph-attention network. Fault-consistent transients transplanted from published source models of California events, and the 2019 Ridgecrest postseismic signal, test the conclusions on realistic signals. First results with simple filters show that using the nearest stations removes about 14% of short-period scatter but keeps only about a quarter of a 25-km-wide transient, and that excluding stations within 50 km keeps most of it at little cost in noise removal. The audit will provide trade-off curves and practical guidance for choosing the reach of a filter, and will establish whether deep networks improve the trade-off or only the scatter.

1 Introduction

Continuous GNSS networks record the positions of thousands of geodetic monuments every day, with formal uncertainties of about 1 mm in the horizontal and 3–5 mm in the vertical. The Nevada Geodetic Laboratory (NGL) processes more than 17,000 such stations worldwide with a single, uniform strategy and distributes the time series openly (Blewitt et al., 2018; Nevada Geodetic Laboratory, 2026). These records contain steady interseismic motion, coseismic offsets, postseismic relaxation, slow slip, magmatic inflation and hydrological loading, together with artefacts from antenna and receiver changes. Separating these contributions is the central task of geodetic time-series analysis.

The main obstacle is noise that is shared across stations. Errors in satellite orbits and clocks, in the realisation of the reference frame and in large-scale environmental loading produce a common-mode component of several millimetres that is coherent over hundreds to thousands of kilometres (Wdowinski et al., 1997; Dong et al., 2006; Kreemer and Blewitt, 2021). A wide

range of filters estimates and subtracts this component, from regional stacking (Wdowinski et al., 1997) through principal and independent component analysis (Dong et al., 2006; Shen et al., 2014; Gruszczynski et al., 2018; Zhou et al., 2022; Liu et al., 2026) to correlation-weighted and robust median filters (Tian and Shen, 2016; Kreemer and Blewitt, 2021). These filters are almost always judged by how much they reduce residual scatter or station-to-station correlation, or by how much they shrink velocity uncertainties. None of these criteria distinguishes noise from signal: a filter that also removes a real transient scores better, not worse.

This matters because the transients of interest are themselves spatially coherent. A slow slip event, afterslip or a hydrological load moves many neighbouring stations together, and a filter that estimates the common mode from those neighbours absorbs it. Kreemer and Blewitt (2021) note that much of the scatter reduction of their method comes from local common-mode transients being absorbed into the estimated common mode. The scientific consequences are measurable: in Cascadia, slow-slip scaling laws, b-values and stress drops all change with the detection threshold applied after denoising (Costantino et al., 2026). The disagreement over a reported precursory acceleration before large earthquakes (Bletery and Nocquet, 2023) turned on the same question, whether a common-mode correction removes only noise or also real deformation (Bradley and Hubbard, 2023; Bletery and Nocquet, 2025).

The most direct attempt to control this loss is that of Bachelot et al. (2025). They train a graph neural network to predict each station from other stations, with the target station masked, and subtract the prediction as noise. The decisive design choice is that each station is connected only to its eight nearest stations more than 400 km away, so that the network sees the common mode but not a slow slip event under the target. The 400 km was chosen empirically, as the shortest distance at which 280 slow-slip offsets from 15 Cascadia events were no longer reduced by denoising. A simple average of the stations beyond 400 km performed about as well as the network, and better on one metric, and small offsets were slightly inflated by denoising while large ones were slightly reduced. Their work shows that excluding nearby stations is the key control on signal loss. It leaves open how the right distance depends on the size of the transient, what any filter actually loses when there is no ground truth for the offsets it is compared against, and how the result transfers to other networks and tectonic settings.

Our own preliminary analyses point to the same question. At eleven continuous stations within 80 km of the 2019 Ridgecrest sequence, every apparent anomaly before the mainshock was explained by network-wide noise, an exceptionally wet winter or equipment changes rather than by deformation on the fault. We then built a leakage-controlled benchmark on 1,251 California stations and ran the first planted-signal tests with two simple filters, a distance-weighted mean and a robust weighted median of neighbouring stations. Using the nearest stations, the weighted mean removes about 14% of short-period scatter but keeps only 27% of a 25-km-wide planted transient, 5% of a 50-km-wide one and 36% of the observed Ridgecrest postseismic displacement. Excluding stations within 50 km keeps 83% of the 25-km-wide transient and 74% of the Ridgecrest signal, while still removing 13% of the scatter. Near the rupture, the filter also created spurious transients: at one station 69 km from the epicentre, the cleaned series showed an opposite-sign signal 2.4 times larger than the real one, because its neighbours closer to the rupture had moved more. The robust median kept the same amount of signal as the mean and removed less noise. These numbers were computed with idealised transients on the test years of the benchmark and will be recomputed under the final protocol (Section 3.7), but they show that the trade-off is large and measurable.

Here we propose a signal-loss audit of network filtering. We test the hypothesis that the signal a filter removes is controlled mainly by the ratio of the transient's footprint to the filter's reach, and that this holds across filter families, from stacking to deep networks. The study has four objectives. First, we measure, with known transients planted in real noise, how much signal each filter retains and how much spurious signal it creates, as a function of transient width, duration and amplitude. Second, we audit the classical filters in common use and the fixed-distance

design of Bachelot et al. (2025) under the same protocol as leave-one-out reconstruction models with a systematically varied exclusion radius. Third, we test whether deep networks improve the trade-off between noise removed and signal kept, rather than scatter reduction alone. Fourth, we confirm the conclusions on realistic, fault-consistent transients and on the Ridgecrest postseismic signal, and translate them into guidance for choosing the reach of a filter.

2 Literature Review

2.1 Classical common-mode filtering

Network filtering of GNSS data began with the Southern California Integrated GPS Network, where the common-mode error was estimated as the weighted mean of the residuals at the other stations and subtracted from each series (Wdowinski et al., 1997). Principal component analysis and the Karhunen–Loève expansion extended this to common modes that vary in space (Dong et al., 2006). Later work made principal component filtering applicable to real networks, in which stations start and stop at different times: a modified analysis that reproduces each series from its principal components without filling gaps (Shen et al., 2014), probabilistic principal component analysis that treats missing days as latent values (Gruszczynski et al., 2018), and a variational Bayesian version that also selects the number of components, tested on 44 stations in southern California (Li et al., 2020). Because the common mode varies in space, partitioning large networks into sub-regions improves on a single global analysis; on 208 Chinese stations, adaptive sub-regional filtering reduced residual scatter by 20–29%, compared with 15–18% for global principal component analysis (Liu et al., 2026). Independent component analysis offers an alternative decomposition (Gualandi et al., 2016; Zhou et al., 2022).

Tian and Shen (2016) weighted each neighbouring station by its distance and by the correlation of its residuals with the target, chose the weights by a grid search that minimises the residual scatter after filtering, and outperformed principal component analysis by more than 13% in scatter reduction on about 1,600 stations in western North America. They also stepped through minimum-distance thresholds to separate long-range common modes from regional ones and to estimate the spatial extent of regional transients. The common-mode component imaging of Kreemer and Blewitt (2021) takes a correlation-weighted median over nearby stations, including the target station itself when it qualifies as a filter station, and reduced residual scatter by 39–50% on about 2,500 stations in western Europe. Across this family, performance is reported as scatter reduction, reduced inter-station correlation, velocity uncertainty or the recovery of a simulated common mode; none of these measures what the filter removes from real transients.

2.2 Distance-excluded and learned filters

Bachelot et al. (2025) are the closest precedent for this study. Their graph neural network represents stations as nodes and connects each to its eight nearest stations beyond 400 km. During training, 30% of the nodes are masked at random and the mean squared error is computed on the masked nodes only, so the network learns to predict a station from distant stations; the prediction is subtracted as noise. They compare it with three stack filters: the stack of Wdowinski et al. (1997), a stack that excludes stations within 50 km of tremor, and a stack of all stations beyond 400 km. On NGL data the network reduced horizontal common-mode noise by more than 70%, and the stacks performed comparably, the 400 km stack better on the network-average metric. Training used a random 80/20 split of time windows for validation, with 2022–2023 held out for testing.

Costantino et al. (2024) take a supervised route. Their network combines graph recurrent layers and transformers, learns its own adjacency between stations, and is trained to extract slow slip signals from synthetic examples: slip on the subduction interface added to noise synthesised from real Cascadia series by principal component decomposition and Fourier phase

randomisation. Validated by synthetic recovery and by agreement with tremor, it was later used to invert fifteen years of denoised data for the slip history of the interface (Costantino et al., 2026). Because it is taught what a slow slip event looks like, it targets signals of that form; a self-supervised filter that only predicts each station from others makes no assumption about the signal. Mastella et al. (2025) model high-frequency scatter at single stations, with noise defined as the residual of a greedy trajectory decomposition (Bedford and Bevis, 2018) and synthetic training noise generated by an adversarial network trained on about 5,000 NGL stations; spatial correlation is deliberately not used. Teutschmann et al. (2026) pretrained a self-supervised foundation model on 17,651 NGL stations by masked latent prediction, also station by station, and fine-tuned it for displacement forecasting and the localisation of seismic offsets.

Predicting a withheld sensor from its neighbours is a familiar principle elsewhere. Tilt and compliance noise is removed from ocean-bottom seismometer records by predicting the vertical channel from the horizontal and pressure channels (Crawford and Webb, 2000; Janiszewski et al., 2019), distributed acoustic sensing records are denoised by predicting each channel from its neighbours (Batson and Royer, 2019; van den Ende et al., 2023), and graph neural networks for spatiotemporal kriging and imputation are trained by reconstructing randomly withheld sensors (Wu et al., 2021; Cini et al., 2022; Marisca et al., 2022). Excluding nearby observations to prevent spatial autocorrelation from inflating apparent skill is likewise established in buffered spatial cross-validation (Le Rest et al., 2014).

2.3 How signal preservation has been assessed

Table 1 summarises how the studies above test whether a filter preserves real signal. Three approaches recur. Observed transients can be compared before and after filtering, as Bachelot et al. (2025) do with slow-slip offsets; this uses real signals but no ground truth, since the reference offsets are themselves noisy and the comparison cannot say how much of a signal was removed. Synthetic signals can be added to synthetic noise, as in Costantino et al. (2024) and Mastella et al. (2025); this provides ground truth but depends on how realistic the synthetic noise is, and has been analysed as a function of signal-to-noise ratio rather than of spatial scale. Independent data such as tremor can confirm that extracted signals are real (Costantino et al., 2024), but cannot measure what was lost. The classical filters are evaluated on noise alone.

2.4 Transient detection with synthetic training data

Labelled geodetic transients are scarce, so the most successful detectors are trained on synthetic signals. Costantino et al. (2023) injected 60,000 synthetic slow slip events, computed with elastic dislocation models (Okada, 1992), into surrogate noise derived from real Cascadia series, and trained a convolutional, attention-based detector on 135 stations. The detector recovered 35 of 40 events in an independent catalogue (Michel et al., 2019) and reached a detection threshold of about M_w 6.2–6.4. Other strategies include training on real labelled windows, which reached about 75% accuracy in southwest Japan (Tanaka et al., 2025), and convolutional autoencoders for InSAR time series (Rouet-Leduc et al., 2021). Classical network-based detectors include the network strain filter (Ohtani et al., 2010), Kalman smoothing with principal component analysis (Ji and Herring, 2013) and sparse estimation (Riel et al., 2014), and the SCEC exercise compared detection methods on synthetic and real southern California data (Lohman and Murray, 2013). In California, the only systematic statewide analysis of transient deformation is the classical study of Klein et al. (2019), which used about 1,000 stations over 1999–2018 and separated postseismic, hydrological, anthropogenic and magmatic signals. Well-documented California transients suitable as templates include the propagating shallow slow slip events of 2023 on the Superstition Hills and Imperial faults (Materna et al., 2024) and the afterslip of the 2010 El Mayor–Cucapah earthquake (Gonzalez-Ortega et al., 2014).

Table 1: How signal preservation has been assessed in common-mode filtering and GNSS denoising.

Study	Test of signal preservation	Ground truth	Dependence on spatial scale
Tian and Shen (2016)	Minimum residual scatter; scatter against distance threshold	None	Transient extent inferred, signal loss not measured
Kreemer and Blewitt (2021)	Scatter reduction, velocity repeatability	None	Absorption of local transients noted, not measured
PCA and ICA filters[a]	Simulated common mode, inter-station correlation, velocity uncertainty	Noise only	Not examined
Mastella et al. (2025)	Synthetic trajectories in synthetic noise	Yes (synthetic)	Single station
Costantino et al. (2024)	Synthetic slow slip in synthetic noise; tremor	Yes (synthetic)	Analysed against signal-to-noise ratio
Bachelot et al. (2025)	280 observed slow-slip offsets before and after	None	One distance chosen (400 km)
This study	Known transients planted in real noise; fault-consistent transplanted events; Ridgecrest postseismic	Yes (planted)	Retention and spurious signal mapped against footprint and exclusion radius

[a] Shen et al. (2014); Gruszczynski et al. (2018); Li et al. (2020); Zhou et al. (2022); Liu et al. (2026).

2.5 Precursory deformation and forecasting

The geodetic evidence for precursory deformation remains contested. The hour-scale acceleration reported by Bletery and Nocquet (2023) from 3,026 high-rate series before 90 earthquakes of $M_w \geq 7$ vanishes after common-mode correction (Bradley and Hubbard, 2023), which the original authors attribute to the correction removing tectonic signal (Bletery and Nocquet, 2025). Stacked tiltmeter records before the 2011 Tohoku-oki earthquake show no precursory signal above a detection limit of about M_w 6.4 (Hirose et al., 2024), and the months-long motions reported before great subduction earthquakes (Bedford et al., 2020) have also been questioned. The best-supported cases are local and slow, such as the eight-month slow slip event before the 2014 Iquique earthquake (Socquet et al., 2017). In earthquake forecasting, geodesy has improved skill mainly through static strain-rate maps combined with smoothed seismicity (Strader et al., 2018; Bayona et al., 2020); afterslip moment does not correlate with aftershock productivity across 41 earthquakes (Churchill et al., 2022), neural forecasting models have not outperformed the epidemic-type aftershock sequence model in California (Stockman et al., 2024), and simple models often match complex networks (Mignan and Broccardo, 2019, 2020). Each of these questions depends on knowing what a filter does to transient signals.

2.6 Synthesis

Five gaps emerge. First, no study has measured with ground truth how much of a transient a common-mode filter removes as a function of the transient's spatial scale relative to the filter's reach. Second, the reach that controls signal loss has been chosen empirically, as a fixed 400 km in Cascadia (Bachelot et al., 2025) or by scatter-based thresholds (Tian and Shen, 2016). Third, filters that use the target station, which include most classical methods, have not been compared with leave-one-out filters on signal loss. Fourth, the biases that filtering introduces have been noted only qualitatively, as the inflation of small slow-slip offsets and reduction of large ones after denoising (Bachelot et al., 2025), and spurious signals created by filtering near large transients have not been quantified. Fifth, deep networks have been compared with simple filters on noise

reduction, not at equal signal retention, and network-aware deep denoisers have been developed only for subduction zones.

3 Proposed Method

Figure 1 summarises the processing chain. Daily positions are reduced to residual displacements by removing each station's long-term trajectory; the residuals are screened and arranged in a single array; each filter then predicts every station from other stations, and the cleaned series are audited against noise, planted transients and real offsets. The following subsections describe each stage.

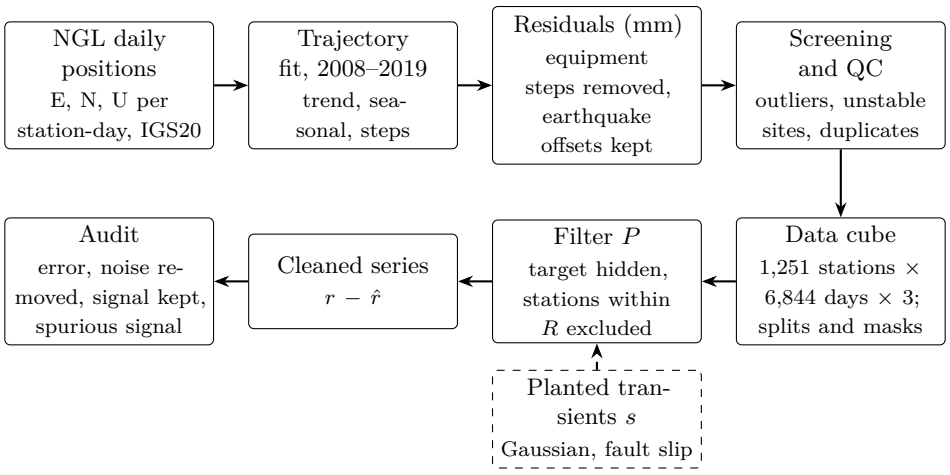


Figure 1: Processing chain from daily GNSS positions to the signal-loss audit. Planted transients are added to the residuals of the data cube, and every filter is run with and without them.

3.1 Data

The data are the daily IGS20 solutions of the Nevada Geodetic Laboratory (Nevada Geodetic Laboratory, 2026). For each station and day, a solution gives the position of the GNSS antenna in metres, in east, north and up components relative to a fixed reference point near the station, estimated from 24 hours of satellite observations, together with formal uncertainties of about 1 mm in the horizontal and 3–5 mm in the vertical. The record is therefore a three-component displacement time series sampled once per day: it resolves periods from two days to decades, far below the band of seismic recordings, and it measures position directly, with no instrument response to remove.

We use all stations between 32.4° and 42.1°N and between 124.6° and 114.0°W with at least 1,095 daily solutions, 1,381 stations in California and adjacent parts of Nevada, Oregon and Baja California. We restrict the analysis to 2008 onward, when network density is approximately constant, and remove 130 stations: 30 with less than one year of data after 2008; 22 with non-tectonic site motion such as groundwater-driven subsidence in the Central Valley, identified by a robust residual scale above 10 mm in the horizontal or 30 mm in the vertical; and 78 co-located duplicates within 1 km of a longer-running station. The remaining 1,251 stations (Figure 2a) provide 6.45×10^6 station-days. In 2018 the median distance to the nearest station was 9.7 km, and a typical station had 23 neighbours within 50 km and 75 within 100 km.

3.2 From positions to residuals

Raw positions are dominated by steady plate motion, tens of millimetres per year in California, and by seasonal loading cycles; the transients of interest are millimetres to centimetres. The

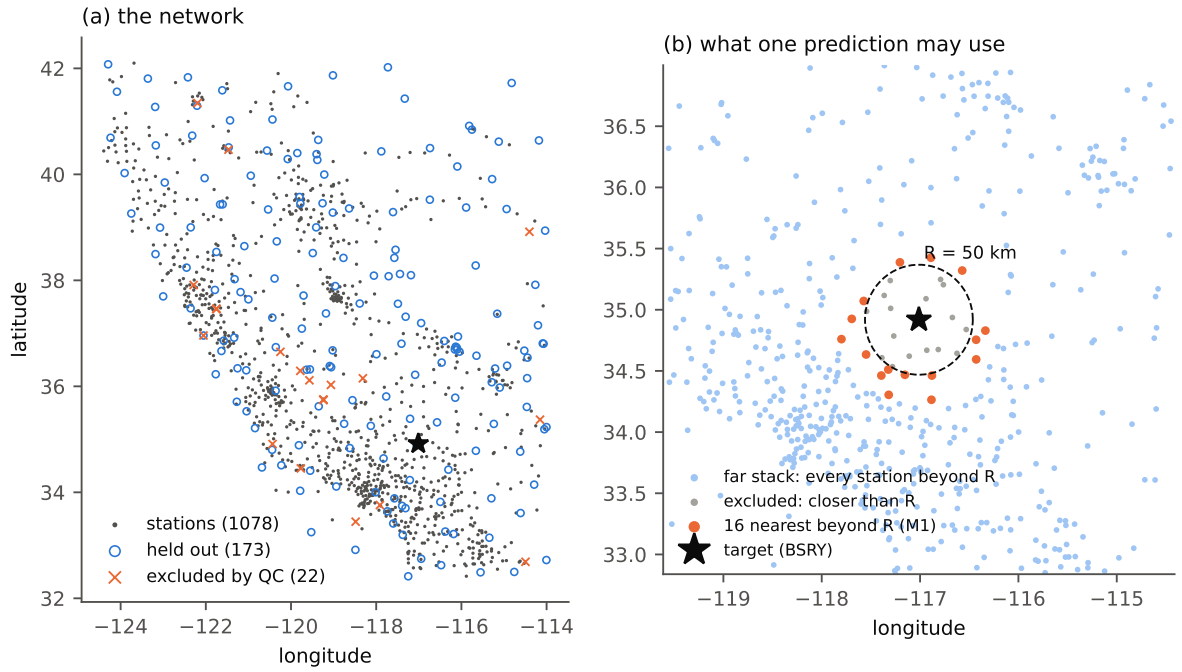


Figure 2: (a) The 1,251 stations used, the 173 held out from all training, and the 22 excluded for non-tectonic site motion. (b) What a prediction for one target station may use at an exclusion radius $R = 50$ km: stations closer than R are never used; a local filter such as the weighted mean uses the 16 nearest stations beyond R; the far stack averages every station beyond R.

positions are reduced to residuals in four steps (Figure 3).

First, for each station i and component $c \in \{E, N, U\}$ we fit by least squares a trajectory model with a constant, a linear rate, annual and semi-annual harmonics and Heaviside steps at the epochs listed in the Nevada Geodetic Laboratory step catalogue,

$$x_{i,c}(t) = a + v\,(t - t_0) + \sum_{k=1}^{2} \left[s_k \sin(2\pi k t) + c_k \cos(2\pi k t) \right] + \sum_j b_j\, H(t - t_j) + r_{i,c}(t), \qquad (1)$$

using the training years 2008–2019 only. The fitted trajectory is then extrapolated to later years, so that no evaluation day influences it.

Second, we subtract the steps at equipment changes, which are artefacts of antenna and receiver replacements; changes after the training years are removed with a local estimate from 30 days on either side. Earthquake offsets are kept in the residuals, because they are real, spatially coherent deformation of the kind a filter should preserve. In Figure 3b the 2010 El Mayor–Cucapah and 2019 Ridgecrest offsets therefore remain in the residual, and the slow motion after Ridgecrest is visible as a drift.

Third, observations more than five robust standard deviations from a 61-day running median are treated as missing, and stations with unstable monuments or non-tectonic motion are removed as described above. Stations that develop site problems after the training years are excluded from scoring.

Fourth, the residuals $r_{i,c}(t)$, in millimetres, are arranged with their formal uncertainties and an availability mask in an array of 1,251 stations by 6,844 days by three components. We assign 2008–2019 to training and 2020–2021 to validation; later years are reserved for reporting and a prospective test. A total of 173 stations, spread across the network, are withheld from all training, and the interval from July 2019 to June 2020 within 150 km of the Ridgecrest rupture is excluded from training, so that its postseismic signal is unseen by every model. Learned models are trained on residuals from which the earthquake offsets estimated in the training fit

have been subtracted, and without the 30 days around each earthquake of magnitude 6 or larger, so that they learn the noise rather than the earthquakes.

We filter displacement residuals rather than daily rates of change. Differencing acts as a high-pass filter: it emphasises day-to-day scatter and suppresses slow signals, so that 2 cm of afterslip spread over a year becomes about 0.05 mm/day against a daily scatter of about 1.5 mm. For transients lasting days to months, the signal-to-noise ratio is highest in displacement. Day-to-day increments are nevertheless given to the attention network as an additional input, because they make the onset of a transient explicit.

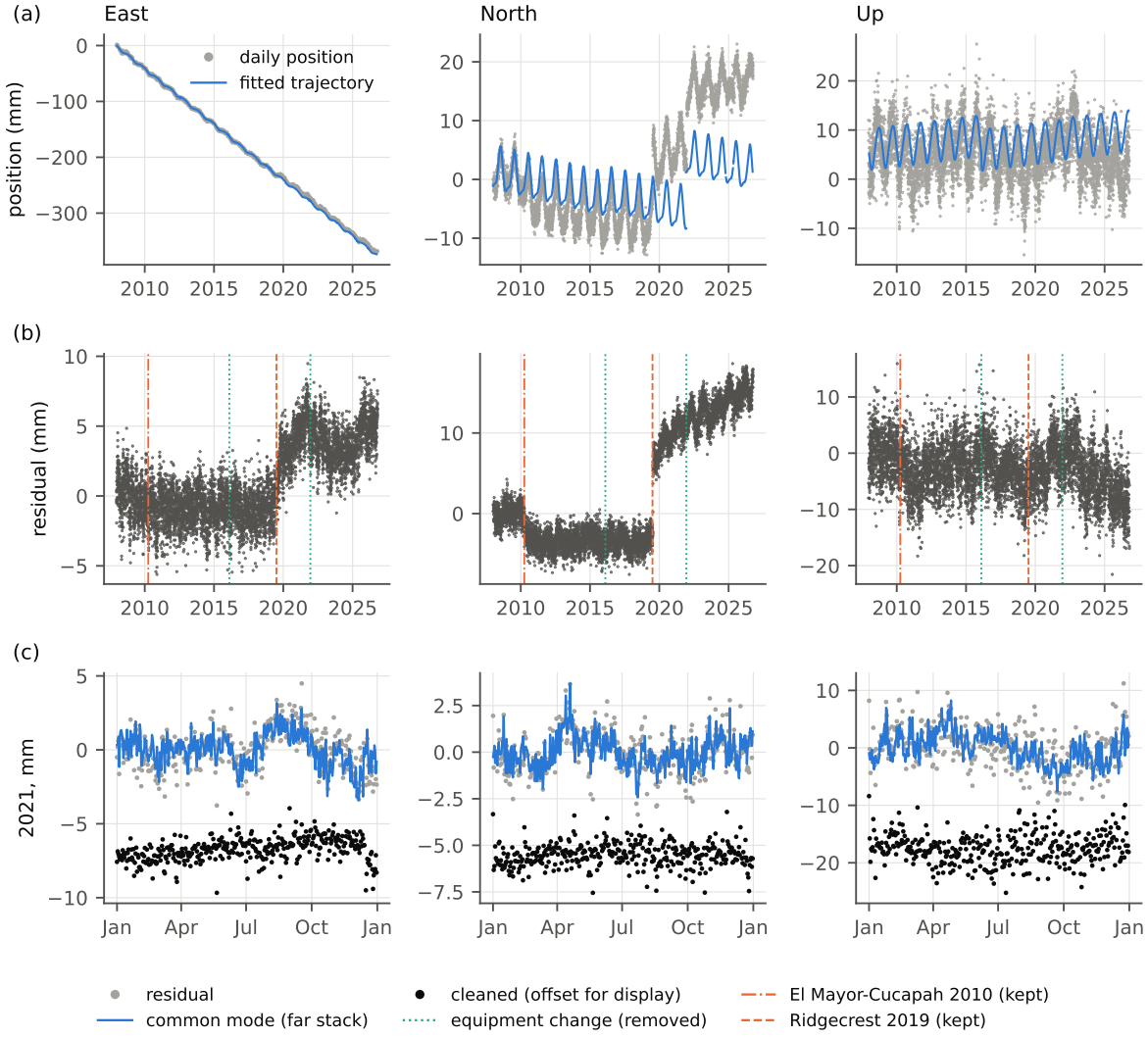


Figure 3: From positions to residuals at station BSRY, about 110 km from the 2019 Ridgecrest earthquakes. (a) Daily positions relative to the first epoch and the trajectory model fitted to 2008–2019; steady plate motion dominates the east component. (b) The residual used by every filter: trend, seasonal terms and equipment steps removed, earthquake offsets kept. (c) One validation year: the residual, the common mode estimated from the rest of the network (the far stack, mean of all other stations) and the cleaned series, shifted down for display. Residuals and common mode are centred on the year's mean.

3.3 From residuals to cleaned series

A filter P estimates the part of each station's residual that is shared with other stations, $\hat{r} = P(r)$, and the cleaned series is $r - \hat{r}$. For the leave-one-out filters, the prediction for a target station

never reads that station's own values, and every station closer than an exclusion radius R is removed from its input (Figure 2b). One pass over the network then cleans every station at once without using it. Classical filters that include the target in their own estimate are audited the same way, under their own names, but are not scored on reconstruction error, where reading the target would be trivially rewarded.

3.4 Signal-loss accounting

The audit rests on four measurements, computed identically for every filter on cleaned series $r - \hat{r}$, where $\hat{r}$ is the filter's estimate of the common mode.

Retention. We plant a transient s of known form in the real residuals and filter both the original and the planted data. With $\Delta = (r + s - \hat{r}_{r+s}) - (r - \hat{r}_r)$ the change in the cleaned series caused by the plant, the retention

$$\rho = \frac{\langle \Delta,\, s \rangle}{\langle s,\, s \rangle} \tag{2}$$

is the fraction of the transient left in the cleaned data, computed over the horizontal components, the transient's days and the stations within twice its width. Pairing the planted and unplanted runs removes the noise term that an unpaired comparison would carry; for a linear filter ρ reduces exactly to $\langle s - \hat{s}, s \rangle / \langle s, s \rangle$. The first set of transients has Gaussian footprints, $s_i(t) = A \exp(-d_i^2 / 2L^2)\, g(t)\, \hat{u}$, with amplitudes A of 2–10 mm, widths L of 10–100 km and durations of 10–90 days: a raised-cosine rise, a 60-day hold and a symmetric fall. Fifty transients per combination, 1,800 in total, are placed at random stations and times and packed into shared runs, separated in space and time so that no station near one transient uses a neighbour near another.

Spurious signal. At stations beyond the footprint of a planted transient whose neighbours lie within it, the planted signal is zero, so any change Δ is created by the filter. We report its peak amplitude as a fraction of A. This quantifies the opposite-sign artefacts we observed near Ridgecrest.

Amplitude dependence. For nonlinear filters, retention can depend on amplitude. Reporting ρ for each A tests the observation of Bachelot et al. (2025) that small offsets were inflated and large ones reduced by denoising.

Noise removed. On the test period, we compute for each station the reduction in short-period scatter, $1 - \mathrm{rms}(\mathrm{fast}(r - \hat{r}))/\mathrm{rms}(\mathrm{fast}(r))$, where the fast part is the series minus its 61-day running median, and take the median over stations.

Plotting noise removed against retention, for each filter and each exclusion radius, gives the trade-off curve that is the central result of the audit. Figure 4 illustrates the measurement for one planted event.

Three further tests use realistic or real signals. In the event library, transients are placed on random faults of the appropriate style in the SCEC Community Fault Model (Plesch et al., 2007, 2024), and the displacement each real station would record is computed with triangular dislocations in an elastic half-space (Nikkhoo and Walter, 2015). Two templates are used, with parameter ranges taken from the full texts of the source papers: shallow slow slip propagating along strike, after the 2023 Superstition Hills and Imperial fault events (Materna et al., 2024; Vavra et al., 2024), and afterslip decaying logarithmically, after the 2010 El Mayor–Cucapah earthquake (Gonzalez-Ortega et al., 2014). Unlike the Gaussian footprints, these reproduce the opposite motion on the two sides of a strike-slip fault, against which a neighbour average partly cancels. Events are kept only where at least three stations move by more than 1 mm; shallow creep triggered on a neighbouring fault (Ramos et al., 2020; Xu et al., 2020) was considered and dropped, because creep confined to the top kilometre is not resolved by the network at this density. Second, every earthquake offset in the step catalogue during the scoring years is compared before and after cleaning: the offset in the raw residual (mean of the ten days after

minus the ten days before) and the share of it left in the cleaned series, for offsets of at least 3 mm with no other catalogued step nearby. Third, we measure the fraction of the observed Ridgecrest postseismic displacement retained at stations within 80 km, on days no model was trained on, and apply the before-and-after offset comparison of Bachelot et al. (2025) to the same signal for direct comparison with their metric. The real offsets lack ground truth, because the observed offset includes noise, and complement rather than replace the planted signals.

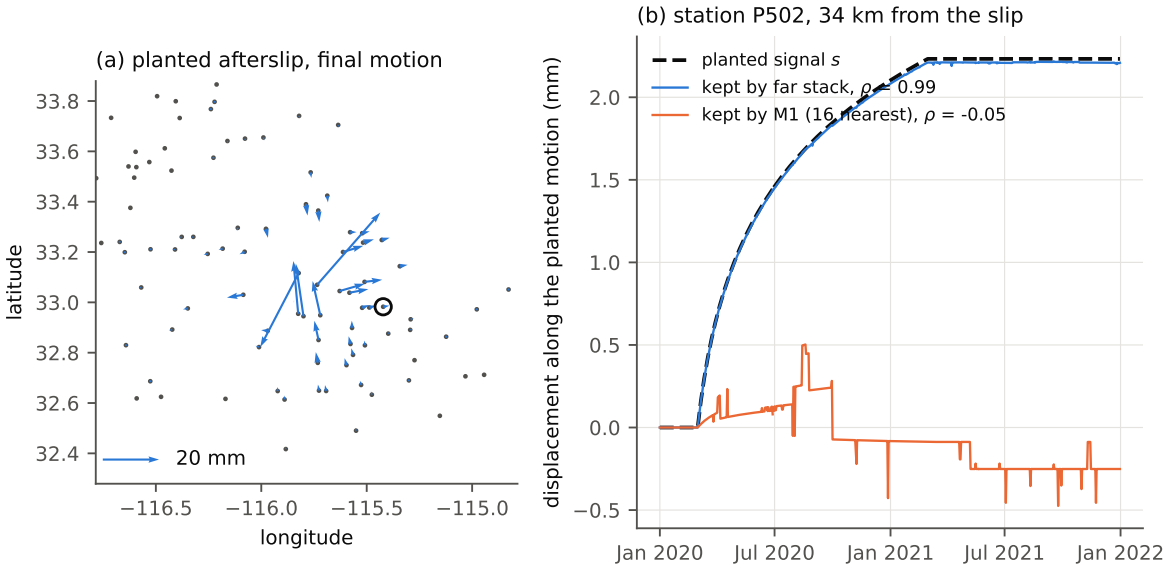


Figure 4: Measuring signal kept. (a) Afterslip planted on a fault of the Community Fault Model: final horizontal displacement at each station, with the slipping patch shaded. (b) At station P502, 34 km from the patch, the planted signal (dashed) and the change it causes in the cleaned series under two filters, projected on the direction of the planted motion; ρ is the retention of Equation (2) at this station. P502 is the station with the median retention under the weighted mean among the 17 stations 15–40 km from the patch. The far stack keeps the signal; the weighted mean of the 16 nearest stations removes it, because its neighbours move with the target.

3.5 Filters audited

Table 2 lists the filters, grouped by how they treat the target station and how their reach is set.

The target-inclusive filters C1–C3 represent common practice: each estimates a station's common mode from data that include the station itself, so a transient local to the station can be absorbed by construction. Because they see the target, they cannot be scored on reconstruction of withheld data; they are scored only on retention, spurious signal and noise removed. D1 reproduces the 400 km stack of Bachelot et al. (2025) on the California network; for their graph neural network we compare with the results they report. The leave-one-out models M1–M5 never see the target station and exclude all stations closer than R, which we vary from 0 to 400 km, so that $R = 400$ km reproduces the fixed distance of Bachelot et al. (2025) as one point on a continuum. M1 and M2 are the simple filters of the preliminary results. M3 learns a weight for each neighbour, in a per-station variant and a variant pooled over stations with distance and azimuth as features, so that it also applies to stations absent from training. M4 lets the weights depend on frequency by regressing band-pass filtered series. M5 is a spatiotemporal transformer trained as a masked autoencoder: a convolutional patch stem embeds a 64-day window at each station, and blocks of divided space-time attention alternate temporal attention within each station with spatial attention across neighbours, biased by distance and azimuth. Its inputs

Table 2: Filters in the audit.

Label	Filter	Reach	Reference
Target-inclusive (classical)			
C1	Regional stack	Whole region	Wdowinski et al. (1997)
C2	Robust weighted median including the target	Nearest stations	Kreemer and Blewitt (2021)
C3	Sub-regional probabilistic PCA	Sub-region	Gruszczynski et al. (2018); Liu et al. (2026)
Fixed distance exclusion			
D1	Mean of stations beyond 400 km	$d > 400$ km	Bachelot et al. (2025)
Leave-one-out with exclusion radius R (this study)			
M1	Distance-weighted mean	$d > R$	Wdowinski et al. (1997)
M2	Robust weighted median	$d > R$	Kreemer and Blewitt (2021)
M3	Ridge regression on neighbours, per station and pooled	$d > R$	Linear spatial prediction
M4	Band-split ridge regression	$d > R$	Multichannel Wiener filter
M5	Space-time attention network; M5-aug trained with planted transients	$d > R$	Bachelot et al. (2025); Costantino et al. (2024)

include day-to-day increments, so that changes are explicit. A variant, M5-aug, adds synthetic transients to the neighbours' inputs during training while leaving the targets unchanged, to test whether a network can learn to ignore transient-shaped shared signal instead of relying on an exclusion radius.

The trainable models are fitted by minimising, over randomly withheld entries $\mathcal{M}$ of the training period,

$$\mathcal{L}(\theta) = \sum_{(i,t,c)\in\mathcal{M}} \frac{(r_{i,c}(t) - \hat{r}_{i,c}(t))^2}{\sigma^2_{i,c}(t) + \sigma_0^2}, \tag{3}$$

with $\sigma_0 = 0.5$ mm. Learned models train on quiet residuals, from which the earthquake offsets estimated in the training-period fit are subtracted; windows within 30 days of catalogued earthquakes are excluded, and residuals are normalised by each station's noise level. A few stations with large coseismic offsets therefore cannot dominate the fit, and the models learn the noise rather than the coherent signal they should leave alone. The same objective drives every learned filter to predict, and therefore remove, any signal its inputs share with the target; this is why the audit measures retention directly rather than relying on the loss. A model that sees the target's own history can also extrapolate a transient's onset into withheld days, so cleaned series and all retention scores are computed with the target withheld for the whole period, and M5 is evaluated with and without its own history.

3.6 Comparison with Bachelot et al. (2025)

Table 3 sets out how the proposed audit relates to the closest prior study. The two share the central idea of masking the target and excluding nearby stations; they differ in what is measured and how the exclusion distance is chosen.

3.7 Evaluation protocol

Every model is scored per station and summarised by the median over stations, so that a few stations with large earthquake offsets cannot dominate. Reconstruction errors on withheld data

Table 3: The design of Bachelot et al. (2025) and of this study.

Aspect	Bachelot et al. (2025)	This study
Region and setting	Cascadia, subduction slow slip	California, strike-slip plate boundary
Target station	Masked	Withheld (leave-one-out)
Exclusion distance	Fixed at 400 km	Swept from 0 to 400 km; 400 km included
How the distance is chosen	Shortest distance that preserves observed slow-slip offsets	Read from measured trade-off curves for each transient width
Test of signal preservation	280 observed offsets before and after denoising	Known planted transients, fault-consistent transplanted events, Ridgecrest postseismic signal
Filtering artefacts	Offset bias noted (small offsets inflated, large reduced)	Retention against amplitude; spurious signals measured
Filters compared	Graph network and three stacks	Target-inclusive, fixed-distance and leave-one-out families, linear to deep

are split into a fast part, the error minus its 61-day running median, and a slow part, and both are normalised by the station's noise level in the training period. Stations that develop site problems after the training period, an increase in day-to-day scatter by more than a factor of three or an unexplained jump, are left out of scoring. All hyper-parameters and design decisions are made on the validation years. Because our first results were computed on 2022–2026 while the scoring rules were still being revised, those years are no longer untouched; we therefore freeze the method at a fixed date and evaluate it once on NGL data recorded after that date as a prospective test. Differences between filters are assessed with a block bootstrap over days and stations, and trainable models are run with several random seeds. We treat tectonic and other aseismic transients as signal and common-mode error and environmental loading as nuisance; daily loading products (Li et al., 2025) allow the loading contribution to the common mode to be estimated separately.

3.8 Extension to continual transient detection

The filter with the best trade-off will provide the residuals for a continual detector: a multi-scale cumulative-sum statistic combined with a spatial-coherence criterion, calibrated to a set number of false alarms per station-year and run in a strictly chronological replay with model updates restricted to windows not flagged as anomalous (Kirkpatrick et al., 2017). This extension depends on the outcome of the audit.

4 Expected Findings

We design the study so that every outcome is informative, and we test four hypotheses.

First, we expect every filter that uses the target station or its nearest neighbours to remove most of any transient wider than the station spacing. In the preliminary results the leave-one-out weighted mean kept about a quarter of a 25-km-wide transient; target-inclusive filters such as stacking, robust median imaging and principal component analysis should keep no more, and less of transients local to a single station, while appearing to remove more noise.

Second, we expect retention to depend mainly on the ratio of transient width to exclusion radius, rising from near zero when the width exceeds the radius to near one when the radius exceeds it, while noise removal declines more slowly because the common mode is coherent over

larger distances. The preliminary results follow this pattern. The 400 km design of Bachelot et al. (2025) should then retain transients up to a few hundred kilometres wide but remove less local noise than intermediate radii; for California, where many transients are a few tens of kilometres wide, we expect a radius of about 25–50 km to give a better balance. Fault-consistent transients should be retained somewhat better than Gaussian ones at the same width, because neighbouring stations on opposite sides of a fault move in opposite directions.

Third, we expect filters to create spurious transients near large real signals, of opposite sign and up to the size of the real signal, at small radii, and these to disappear once the radius exceeds the signal's extent.

Fourth, we expect learned models to reduce reconstruction error relative to the weighted mean, but to improve the trade-off between noise removed and signal kept only modestly, if at all. Consistent with Bachelot et al. (2025), where a simple stack matched a graph network, a deep model that wins only on scatter would be absorbing more signal, and the audit would show it.

The main deliverables are a signal-loss table and trade-off curves for the filter families in common use, measured with ground truth on a dense strike-slip network; a quantitative basis for choosing the reach of a filter for transients of a given size; measured rates of spurious signals; and an open benchmark with data, splits, planted signals and code. Whatever the ranking of the filters, these results will let geodesists report what their filtering costs in signal alongside what it removes in noise.

AI Usage Declaration

An AI assistant, Claude Code running the Claude Opus 5.5 model (Anthropic), was used to help design the benchmark, write the software and its tests, run the preliminary analyses and produce the figures, search and summarise the literature, and draft and edit this text. Bibliographic entries were generated from Crossref and arXiv metadata. The research question and the central idea of predicting withheld stations from their neighbours are the author's. The author directed the work, reviewed all AI-generated material and is responsible for the content. Some cited papers have so far been assessed from their abstracts and will be read in full before any results are submitted for publication.

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